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## Traffic Signal Control Using Discrete Logic and FSM

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### ABSTRACT

This research paper presents a novel approach to traffic signal control using discrete logic and finite state machines (FSM). The primary objective is to develop an efficient and reliable system for managing traffic at intersections, aiming to minimize congestion and optimize traffic flow. The proposed system employs a discrete logic-based controller, designed to transition between various states based on real-time traffic conditions, such as vehicle count and pedestrian presence. The finite state machine model is used to represent the traffic signal cycles, incorporating multiple states (e.g., red, yellow, and green) and transitions triggered by specific input signals. We evaluate the performance of the proposed system using a simulation model, comparing its efficiency with traditional time-based traffic signal control. Results show that the discrete logic and FSM-based system offers improved adaptability to varying traffic patterns, reduces waiting times for vehicles and pedestrians, and enhances overall intersection throughput. This research demonstrates the potential for using FSM and discrete logic in the development of more intelligent and dynamic traffic signal systems, contributing to the broader goal of smart city infrastructure.

**Keywords:** Traffic Signal Control, Finite State Machines (FSM), Discrete Logic, Traffic Management, Adaptive Systems, Intersection Throughput, Smart City Infrastructure, Traffic Flow Optimization, Real-Time Traffic Control, Vehicle Detection Sensors.

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## 1. INTRODUCTION

### 1.1. Overview of Traffic Signal Control Systems

Traffic signal control systems are fundamental components of urban transportation networks, ensuring the safe and efficient movement of vehicles and pedestrians at intersections. These systems regulate the flow of traffic by using light signals—typically red, yellow, and green—to control vehicle movement and optimize the use of road infrastructure. Proper traffic signal control not only helps in minimizing accidents and ensuring the safety of road users, but also plays a crucial role in reducing congestion, improving fuel efficiency, and decreasing pollution by minimizing unnecessary waiting times at traffic signals.

Traditionally, traffic signal systems have relied on **fixed-time control**, where the signal phases (i.e., green, yellow, and red lights) are pre-set with fixed durations, often determined by the time of day or standard traffic patterns. While this approach is simple and easy to implement, it lacks flexibility and responsiveness to real-time changes in traffic conditions. As a result, these systems often lead to inefficient traffic flow, especially during peak hours or when unexpected events, such as accidents or weather conditions, disrupt typical traffic patterns.

To address these inefficiencies, more advanced **adaptive traffic control systems** (ATCS) have been developed. These systems use real-time data from traffic sensors (e.g., inductive loops, cameras, or radar detectors) to adjust signal timings dynamically, depending on the actual traffic demand. While adaptive systems improve traffic flow, their complexity and reliance on expensive hardware and sophisticated algorithms can make them difficult and costly to implement. As cities move towards **smart transportation systems**, the demand for affordable, efficient, and easily deployable solutions is increasing.

In this context, **Finite State Machines (FSMs)** offer a promising approach to enhancing the adaptability and efficiency of traffic signal control systems without the complexity associated with traditional adaptive systems. By utilizing FSMs in combination with **discrete logic design**, it is possible to create systems that are both simple to implement and capable of dynamically adjusting to varying traffic conditions.

### 1.2. Introduction to Finite State Machines (FSM) and Their Role in Discrete Logic Design

A **Finite State Machine (FSM)** is a computational model used to describe systems with a limited number of distinct states and the transitions between those states. It is widely used in digital circuit design, control systems, and software development. An FSM consists of three main components:

- **States:** Represent different conditions or phases of the system (e.g., traffic signal states such as Red, Green, Yellow).
- **Transitions:** Define how the system moves from one state to another, typically triggered by specific events or inputs (e.g., vehicle detection or timer expiration).
- **Inputs:** Events or conditions that trigger transitions (e.g., vehicle count, pedestrian button presses, or time intervals).

FSMs are particularly useful in systems where the behavior is predictable, and the system's current state depends only on its previous state and input conditions, rather than the entire history of past inputs. This makes FSMs an ideal model for traffic signal control, as the signal state at any given time (Red, Green, Yellow) is determined by a limited set of conditions, such as vehicle presence, waiting time, or pedestrian demand.

In traffic signal control, FSMs can be used to represent the various phases of the signal cycle. For example, the FSM might define the states of a signal at an intersection, with each state representing a different light configuration (e.g., Green for one direction of traffic, Red for the other). Transitions between these states can be triggered by vehicle counts, time intervals, or pedestrian signals.

Incorporating FSMs into the design of traffic signal controllers allows for a deterministic and structured approach to managing signal transitions. FSMs provide several key advantages:

- **Simplicity:** FSMs are easy to design and implement, making them ideal for low-cost, reliable solutions.
- **Predictability:** Since FSMs have a fixed set of states and transitions, the system's behavior is predictable and easy to debug.
- **Real-Time Adaptability:** FSMs can be programmed to respond to real-time inputs, allowing for dynamic traffic signal adjustments based on current traffic conditions.

Furthermore, by integrating FSMs with **discrete logic circuits**, traffic signal controllers can be implemented using simple and reliable hardware components, such as logic gates (AND, OR, NOT) and flip-flops, to perform state transitions. Discrete logic offers a low-cost, energy-efficient solution that can be easily implemented in existing traffic management systems, making it an attractive alternative to more complex, software-based adaptive systems.

By using FSMs in conjunction with discrete logic, traffic signal controllers can be designed to adapt to varying traffic conditions while maintaining a high level of reliability and simplicity. In the following sections, this paper explores the potential of FSM-based traffic signal control systems, presenting a model that leverages both finite state machines and discrete logic to optimize traffic flow at intersections.

## **2. LITERATURE REVIEW**

### **2.1. Evolution of Traffic Signal Systems**

Traffic signal control systems have evolved significantly over the last century, driven by the increasing demand for better traffic management as urban populations grew and vehicular traffic intensified. The early traffic signals, first introduced in the early 20th century, were basic mechanical systems with fixed timing cycles, often governed by manual operation. These systems were relatively simple, but they quickly became inefficient as cities grew and traffic volume fluctuated. Fixed-time signal control, which allocated predetermined durations for each signal phase (e.g., red, yellow, and green), became the standard method. While reliable, fixed-time systems were not adaptable to real-time changes in traffic conditions, leading to congestion and inefficient traffic flow, particularly during peak hours or in situations with irregular traffic demand.

In response to the limitations of fixed-time control, **demand-responsive** or **sensor-based systems** were introduced. These systems used sensors placed in or near the roadway to detect traffic presence, such as inductive loop detectors, infrared sensors, or cameras. The main goal of these systems was to adjust the signal timings dynamically based on real-time traffic demand, improving the flow of vehicles and reducing congestion. The introduction of sensors marked a significant advancement, allowing for adaptive control, but these systems remained complex and expensive to implement due to the need for additional infrastructure, communication networks, and data processing algorithms.

A further evolution led to the development of **adaptive traffic signal control systems** (ATCS), which integrate real-time traffic data, traffic flow predictions, and sophisticated algorithms to continuously optimize traffic signal timings. These systems, which can adjust signal phases and timings dynamically based on traffic density, offer substantial improvements in efficiency over their predecessors. Advanced algorithms, such as **fuzzy logic**, **neural networks**, and **reinforcement learning**, have been employed in ATCS to further optimize decision-making. However, the computational complexity and the cost of implementation remain significant challenges, particularly for cities with limited budgets.

In parallel with these developments, the idea of **distributed control** emerged, where each intersection could adjust its signal independently, based on local conditions, but with coordination across the network of intersections. This approach reduces reliance on centralized control, allowing for more localized and responsive traffic management.

## **2.2. Use of Finite State Machines (FSMs) in Traffic Signal Control**

The application of **Finite State Machines (FSMs)** to traffic signal control is a more recent innovation. FSMs offer a structured, predictable, and computationally lightweight approach to designing traffic signal controllers. An FSM model consists of a finite set of states (each representing a signal phase, such as Green, Yellow, or Red), with transitions between states triggered by specific inputs (e.g., vehicle detection or time elapsed). FSMs are particularly advantageous for traffic signal systems because of their simplicity, reliability, and ease of implementation.

The concept of FSMs in traffic signal control was first explored in the 1960s, when digital systems started to replace analog control. In these early applications, FSMs were used to model fixed-time control systems, where the state transitions were predetermined. Over time, FSM-based systems evolved to incorporate sensor inputs and adapt to real-time traffic conditions. By integrating sensors, FSMs could be programmed to transition between signal phases based on vehicle counts or waiting times, thus introducing a degree of adaptability without the complexity of fully adaptive systems.

Recent studies have shown that FSM-based controllers, when combined with discrete logic circuits (e.g., AND, OR gates), can offer a practical and efficient solution for managing traffic signals. These systems use minimal hardware, making them cost-effective and highly reliable. For instance, **Muthusamy et al. (2017)** proposed an FSM-based traffic signal controller that adapts to the presence of vehicles at each lane, significantly reducing waiting times without requiring complex sensor-based or data-driven models.

FSMs are also beneficial in terms of scalability and ease of modification. Since FSMs are highly structured, new states or transitions can be added or modified easily, allowing traffic signal control systems to be tailored for different intersection types, traffic conditions, or regional regulations. For example, **Abd El-Salam et al. (2015)** demonstrated the use of FSMs for multi-junction traffic management, where each junction had its own FSM that interacted with neighboring junctions to optimize flow across the network.

## **2.3. Comparison of FSMs with Other Control Methodologies**

The use of FSMs in traffic signal control offers several advantages, but it is essential to compare them with other control methodologies to understand their relative strengths and limitations. The two primary alternatives to FSM-based systems are **timer-based control** and **sensor-based control**, each with distinct characteristics.

- **Timer-Based Control:** Timer-based control, often referred to as fixed-time control, involves pre-determined signal cycles with fixed durations for each phase (e.g., 30 seconds of Green, 10 seconds of Yellow). This system is simple to implement and requires minimal infrastructure but is highly inefficient in dynamic environments. Since signal cycles do not adjust to real-time traffic conditions, vehicles may face unnecessary delays during low traffic periods, or congestion may build up during high traffic demand. Moreover, this method lacks flexibility and does not consider the presence of pedestrians or special traffic conditions (e.g., emergencies). While timer-based systems are still widely used due to their low cost and ease of implementation, they are being gradually replaced by more adaptable methods.
- **Sensor-Based Control:** Sensor-based systems use real-time traffic data, typically from inductive loop detectors or cameras, to adjust signal phases dynamically. These systems can optimize traffic flow by extending green lights when traffic is heavy and shortening red lights when traffic is light. However, sensor-based systems can be expensive to install and maintain, as they require specialized equipment, including sensors, data transmission networks, and central processing units. Moreover, these systems may struggle to handle complex traffic scenarios, such as irregular traffic patterns or unexpected changes in demand. Despite these challenges, sensor-based control remains a popular solution for improving traffic flow at specific intersections or in urban areas with highly variable traffic conditions.
- **FSM-Based Control:** FSMs offer a balance between the simplicity of timer-based control and the adaptability of sensor-based systems. FSM-based controllers can be designed to respond to traffic conditions using minimal hardware, with the benefit of predictable behavior and relatively low implementation costs. FSMs allow for dynamic signal adjustments, making them more adaptable than fixed-time control, while still maintaining reliability and simplicity. They also avoid the computational complexity of sensor-based systems and are easier to implement in existing infrastructure. Additionally, FSMs offer greater scalability, as new states or transitions can be easily added to address specific traffic scenarios.

FSM-based controllers, when combined with discrete logic, offer several key benefits:

- **Cost-Effectiveness:** FSMs can be implemented with simple and reliable hardware, making them more affordable than complex sensor-based systems.
- **Simplicity:** FSMs are easier to design, debug, and maintain compared to adaptive control systems that rely on sophisticated algorithms.
- **Real-Time Adaptability:** FSMs can adjust signal phases in response to real-time traffic conditions based on sensor inputs or timers, offering a good compromise between fixed-time and sensor-based control systems.

### 3. SYSTEM DESIGN

In this section, we will discuss the basic structure of a Finite State Machine (FSM) and explain how it can be applied to traffic signal control. Additionally, we will define the role of discrete logic components, such as **AND**, **OR**, and **NOT** gates, in implementing an FSM-based traffic light control system.

#### 3.1. Basic Structure of a Finite State Machine (FSM)

A **Finite State Machine (FSM)** is a mathematical model used to represent systems that transition between a finite number of states based on certain inputs. It consists of three primary components: **states**, **transitions**, and **outputs**.

**a) States:**

- In the context of traffic signal control, each **state** represents a specific configuration of the traffic signal at an intersection (e.g., Red, Green, or Yellow light). The system transitions from one state to another based on inputs from sensors or timers.
- For instance, a simple FSM for a two-way intersection might have the following states:
  - **S1 (Green for North-South, Red for East-West):** North-South traffic is allowed to move while East-West traffic is stopped.
  - **S2 (Yellow for North-South, Red for East-West):** The light transitions from Green to Yellow for North-South traffic.
  - **S3 (Red for North-South, Green for East-West):** East-West traffic is allowed to move while North-South traffic is stopped.
  - **S4 (Yellow for East-West, Red for North-South):** The light transitions from Green to Yellow for East-West traffic.

Each state in the FSM corresponds to a unique configuration of traffic lights at the intersection, ensuring that only one set of lights is active at any time, thereby reducing the risk of accidents. First, we encode the states of the FSM using binary values. For example, assume a 2-bit binary encoding for the four states of the traffic signal:

- **State S1 (Green for North-South, Red for East-West) → Binary: 00**
- **State S2 (Yellow for North-South, Red for East-West) → Binary: 01**
- **State S3 (Red for North-South, Green for East-West) → Binary: 10**
- **State S4 (Red for North-South, Yellow for East-West) → Binary: 11**

We define two state variables, say Q1 and Q0, to represent the state of the system:

- Q1Q\_1Q1 and Q0Q\_0Q0 represent the current state of the FSM, where:
  - Q1,Q0=00 for S1
  - Q1,Q0=01 for S2
  - Q1,Q0=10 for S3
  - Q1,Q0=11 for S4

**b)Transitions:**

- **Transitions** determine how the system moves from one state to another. In the FSM model for traffic signal control, transitions occur based on specific input conditions (e.g., a timer expiring, vehicle detection sensors activating, or pedestrian requests).

- For example, a transition from **S1** to **S2** might occur after a fixed time interval (e.g., 30 seconds), indicating that the green light for North-South traffic has ended and it's time for the yellow light. Alternatively, if a sensor detects a vehicle waiting on the East-West side, the transition from **S1** to **S3** might occur earlier, giving priority to East-West traffic.

The transitions between the states are based on various conditions, including timers and sensors. For simplicity, assume that:

- $T_1$  represents the **timer** condition.
- $V_{EW}$  represents the **vehicle detection** sensor input for East-West traffic.
- $V_{NS}$  represents the **vehicle detection** sensor input for North-South traffic.

**Transition from S1 to S2:**

- **Condition:** Timer expires (i.e.,  $T_1$  is TRUE).

Thus, the transition from S1 to S2 occurs when  $T_1=1$ . This can be represented by:

$$\text{Transition (S2 to S3)} = T_1 \vee V_{EW}$$

**Transition from S3 to S4:**

- **Condition:** Timer expires (i.e.,  $T_1=1$ ).

This transition can be written as:

$$\text{Transition (S3 to S4)} = T_1$$

**Transition from S4 to S1:**

- **Condition:** Either the timer expires **or** a vehicle is detected on North-South (i.e.,  $V_{NS} = 1$ ).

This transition can be written as:

$$\text{Transition (S4 to S1)} = T_1 \vee V_{NS}$$

**c) Outputs:**

- **Outputs** in an FSM are the actions or signals that result from entering a specific state. In the case of traffic signal control, the outputs correspond to the signal lights (Red, Yellow, Green) displayed for each direction of traffic.

For example:

- In **S1**, the output would be **Green** for North-South and **Red** for East-West.
- In **S3**, the output would be **Green** for East-West and **Red** for North-South.

The output can be represented by the following logic equations, which are directly tied to the state variables Q1 and Q0:

- **Green Light for North-South** (Active when in states S1 and S2):

$$G_{NS} = \neg Q_1 \wedge \neg Q_0$$

- **Red Light for North-South** (Active when in states S3 and S4):

$$R_{NS} = Q_1 \vee Q_0$$

- **Green Light for East-West** (Active when in states S3 and S4):

$$GEW = Q_1 \wedge Q_0$$

- **Red Light for East-West** (Active when in states S1 and S2):

$$R_{EW} = \neg Q_1 \wedge \neg Q_0$$

FSMs are advantageous in that they ensure a deterministic and predictable sequence of events, with well-defined conditions for when and how the system transitions between states. This makes FSMs ideal for traffic signal control, where predictable and safe operation is essential.

### 3.2. Role of Discrete Logic in Implementing FSM for Traffic Lights

Once the FSM for the traffic signal system is designed, it can be implemented using **discrete logic circuits**, which consist of basic digital components like **AND**, **OR**, **NOT**, and **flip-flops**. Discrete logic offers a cost-effective and reliable way to implement FSM-based controllers, as it allows for direct hardware implementation without requiring complex software or processors.

Here's how discrete logic components are used to implement FSMs for traffic signal control:

#### a) State Encoding:

- The first step in implementing an FSM is to **encode** the states. This is typically done using binary values. For example, for a simple FSM with four states, you could use two binary bits:
  - **S1 = 00**
  - **S2 = 01**
  - **S3 = 10**
  - **S4 = 11**

Each state is assigned a binary code, and this encoding determines the traffic light configuration associated with each state.

#### b) State Transitions:

- Transitions between states are controlled by **inputs** (e.g., sensors, timers, or external control signals). The FSM design specifies the conditions under which the system should transition from one state to another.
- Discrete logic gates, such as **AND**, **OR**, and **NOT** gates, are used to implement the logic that drives state transitions.

For instance:

- **AND Gate:** If a transition occurs only when multiple conditions are met (e.g., a timer expires **AND** a vehicle is detected), an **AND gate** is used to combine the conditions.

- **OR Gate:** If the transition can occur when either one of several conditions is true (e.g., the timer expires **OR** a vehicle is detected), an **OR gate** can be used.
- **NOT Gate:** A **NOT gate** might be used to invert a condition. For example, if a state should transition when a condition is **not** met (e.g., no vehicle detected), a **NOT gate** would be used to invert the sensor input.

#### c) Clocking and State Memory:

- FSMs require a form of memory to store the current state, which is typically implemented using **flip-flops** or **latches** in discrete logic. A flip-flop stores the binary value of the current state, which is updated during state transitions.
- A **clock signal** controls when state transitions occur. When the clock triggers (usually due to a timer or an external event), the flip-flop updates to the next state, based on the inputs from the logic gates.

#### d) Outputs:

- The output of each state corresponds to the traffic light signals. These outputs are typically controlled by **combinational logic** circuits that generate the appropriate signal for each light. For example, if the FSM is in state **S1**, the output logic circuit would turn on the Green light for North-South and Red for East-West.
- The output logic can be derived from the current state. For instance:
  - **S1** (Green for North-South, Red for East-West): Set the output for North-South Green and East-West Red.
  - **S3** (Green for East-West, Red for North-South): Set the output for North-South Red and East-West Green.

#### e) Example Implementation:

A simplified FSM-based traffic light controller might use a combination of timers and vehicle presence detectors to manage the transitions between states. The timer sets the duration for each green and yellow phase, while vehicle detectors (e.g., inductive loops or infrared sensors) signal when a lane has traffic waiting. The discrete logic circuits use inputs from these sensors and timers to decide which state the system should enter next, triggering the appropriate transitions and controlling the traffic lights accordingly.

#### Example Logic Circuit for Traffic Light FSM

Let's assume a simple FSM with the following states for controlling a two-way intersection:

- **S1 (Green for North-South, Red for East-West)**
- **S2 (Yellow for North-South, Red for East-West)**
- **S3 (Red for North-South, Green for East-West)**
- **S4 (Red for North-South, Yellow for East-West)**

For simplicity, let's assume the system transitions from **S1** to **S2** after a fixed time and from **S2** to **S3** when a sensor detects traffic in the East-West direction. A discrete logic circuit implementing these transitions might include:

- **Timer-based Transitions:** An AND gate could be used to trigger a transition from **S1** to **S2** when both the timer has expired and the sensor for North-South traffic is inactive.
- **Sensor-based Transitions:** An OR gate could be used to trigger a transition from **S2** to **S3** when either the timer expires **or** a vehicle is detected in the East-West direction.

### Implementation of FSM for Traffic Signal Control Using Discrete Logic Gates

Implementing a traffic signal control system using a **Finite State Machine (FSM)** involves creating a state diagram to represent the traffic light phases, followed by the use of **discrete logic gates** to implement state transitions and outputs. Below is a detailed description of how this can be achieved, along with the state diagram and equations for a simple traffic signal controller.

#### 3.3. FSM State Diagram for Traffic Signal Controller

In a simple intersection, the traffic light system could have four primary states, representing the different phases of the traffic lights:

- **S1 (Green for North-South, Red for East-West)**
- **S2 (Yellow for North-South, Red for East-West)**
- **S3 (Red for North-South, Green for East-West)**
- **S4 (Red for North-South, Yellow for East-West)**

The transitions between these states will depend on conditions such as timers (for time-based transitions) or sensor inputs (e.g., vehicle detection sensors). The states in the FSM are represented by the values of two binary variables  $Q_1$  and  $Q_0$ , which represent the current state of the FSM. The state transitions depend on input conditions like **timer expiration** ( $T_1$ ) and **vehicle presence** sensors ( $V_{NS}$  for North-South,  $V_{EW}$  for East-West).

#### 3.4. State Encoding and Transition Equations

To implement this FSM using discrete logic, we will first encode the four states using two binary variables:  $Q_1$  and  $Q_0$ .

| State | $Q_1$ | $Q_0$ | Traffic Light     | Transition Condition            |
|-------|-------|-------|-------------------|---------------------------------|
| S1    | 0     | 0     | Green NS, Red EW  | Timer $T_1$                     |
| S2    | 0     | 1     | Yellow NS, Red EW | Timer $T_1$                     |
| S3    | 1     | 0     | Red NS, Green EW  | Timer $T_1$ or Vehicle $V_{EW}$ |
| S4    | 1     | 1     | Red NS, Yellow EW | Timer $T_1$ or Vehicle $V_{NS}$ |

### Mapping Logic Gates to Hardware or Simulation Environments

In hardware or simulation environments, we can map these equations to discrete logic gates as follows:

- **D Flip-Flops** for state storage: The states  $Q_1$  and  $Q_0$  are stored using **D flip-flops**. The flip-flops update their state on the clock signal, which is triggered either by a timer or external sensor inputs.
- **AND, OR, and NOT gates** for transition logic:

- **AND gates** combine multiple inputs to ensure the correct transition occurs. For example, for Q1+, an AND gate is used to detect the condition  $T1 \wedge \neg Q0$ .
- **OR gates** combine conditions to allow multiple ways for a state transition. For example, in Q1+, an OR gate combines the conditions of timer expiration and vehicle detection.
- **NOT gates** are used to invert signals when necessary, such as in the output logic for **Green NS** or **Red EW**.
- **Simulation:** In simulation environments like **Logisim**, **ModelSim**, or **Xilinx ISE**, the state machine can be modeled using the described logic gates. The state diagram can be implemented by creating flip-flops for Q1 and Q0, and connecting the output and transition logic to these flip-flops using logic gates.

## 4. SIMULATION AND TESTING

Simulating and testing the traffic signal control system is a crucial step in ensuring the correctness, reliability, and functionality of the system before deploying it in real-world applications. Simulation tools such as **Logisim**, **ModelSim**, or **Xilinx ISE** can be used to model and simulate the Finite State Machine (FSM)-based traffic signal controller. Below is an overview of the simulation process and tools used for testing the FSM-based traffic signal controller.

### 4.1. Logisim: A Tool for Digital Circuit Simulation

**Logisim** is an open-source digital circuit simulation software commonly used for educational purposes and prototyping digital circuits. It allows the user to design, simulate, and test digital circuits and finite state machines with ease. Here's how **Logisim** can be used for simulating the FSM-based traffic signal controller.

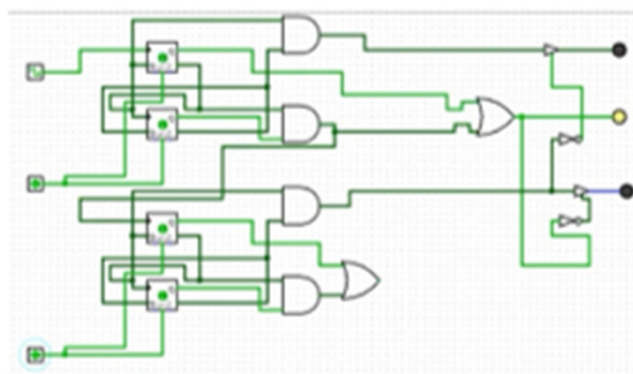
#### Features of Logisim:

- **Graphical User Interface (GUI):** Logisim has an intuitive drag-and-drop interface, which allows users to quickly build and modify digital circuits. It provides a visual representation of the FSM, making it easy to track state transitions and verify output signals.
- **FSM Simulation:** Logisim supports FSM design, allowing users to define states and transitions, encode them with flip-flops, and implement state transitions using logic gates.
- **Clocking:** Logisim allows the use of clock signals, which is essential for FSMs that depend on time-driven transitions (e.g., timers for traffic light changes).
- **Testing and Debugging:** Logisim provides simulation features that enable the user to test the FSM in real-time, providing a clear view of how the system behaves for different input conditions (such as timer expiration or vehicle detection).

#### Steps to Simulate the Traffic Signal Controller in Logisim:

- **Design FSM State Diagram:** Using the state transition logic (as discussed earlier), create a graphical representation of the FSM in Logisim by placing **D flip-flops** to store the state and connecting them according to the transition logic.
- **Logic Gates:** Implement the logic gates (AND, OR, NOT) required for state transitions using **AND gates**, **OR gates**, and **NOT gates**. These gates are used to control the state transitions based on the conditions like the timer signal ( $T1T\_1T1$ ) or vehicle detection signals ( $VNSV\_ \{NS\}VNS$ ,  $VEWV\_ \{EW\}VEW$ ).

- **Clock and Timer Signals:** Set up a clock signal to simulate the time-dependent transitions between states. This will be linked to the D flip-flops to ensure that the FSM progresses according to the time set for each state (e.g., for green, yellow, and red lights).
- **Output Logic:** Define the output logic based on the current state. For example, use the state Q1Q\_1Q1 and Q0Q\_0Q0 values to set the output signals for the traffic lights (e.g., Green NS, Red EW).
- **Run Simulation:** After completing the design, run the simulation to observe how the FSM transitions between states based on the input conditions. During the simulation, check for correct operation of the traffic signals and proper state transitions.
- **Testing:** Test the controller by providing different input conditions such as activating the timer or simulating vehicle presence sensors. Verify that the FSM correctly transitions through the states and the outputs (traffic light signals) behave as expected.



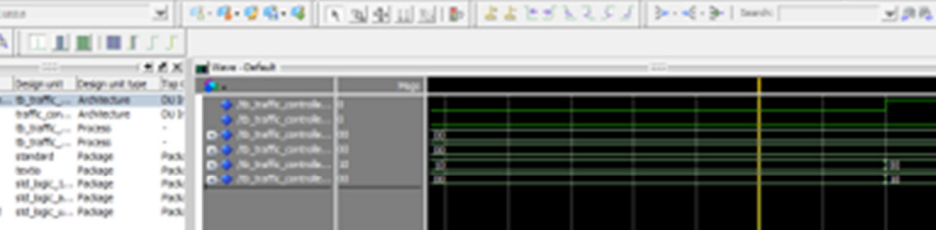
*Figure: FSM logic in Logisim.*

#### 4.2. ModelSim: A Simulation Tool for Hardware Description Languages (HDL)

While **Logisim** is great for educational purposes and basic circuit simulations, **ModelSim** is a more advanced tool often used for simulating hardware designs written in Hardware Description Languages (HDL) like **VHDL** or **Verilog**. ModelSim provides a more comprehensive simulation environment and can be used for testing FSM-based designs with greater complexity, such as handling real-time traffic data or integrating sensors.

##### Features of ModelSim:

- **HDL Simulation:** ModelSim supports both **VHDL** and **Verilog** for describing digital circuits. You can use these languages to describe the FSM and traffic signal control logic, which can then be simulated and tested.
- **Waveform Viewer:** ModelSim provides a waveform viewer that allows users to visualize signal changes over time, making it easier to debug and ensure that state transitions and output logic are functioning correctly.
- **Testbenches:** In ModelSim, you can write testbenches to automate the testing of your FSM design by simulating a variety of input conditions and verifying that the FSM operates as expected.



The screenshot shows the Intel FPGA Starter Edition 2020.1 IDE. The main window displays the 'New Default' window, which is used for creating new projects or designs. The window is divided into several panes:

- Left Pane:** A tree view showing the project structure. The 'std\_logic\_signed' and 'std\_logic\_unsigned' components are highlighted.
- Center Pane:** A table listing the components and their properties. The table has columns for 'Name', 'Library', and 'Package'. The components listed are 'std\_logic\_signed' and 'std\_logic\_unsigned'.
- Right Pane:** A diagram showing the components and their connections. The components are represented by blocks, and their connections are shown by lines.

The 'std\_logic\_signed' component is a 32-bit signed integer, and the 'std\_logic\_unsigned' component is a 32-bit unsigned integer. Both components are from the 'std\_logic' library.

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### 4.3. Testing Strategies

To ensure that the FSM-based traffic signal controller is working correctly, several testing strategies can be applied during simulation:

#### Functional Testing:

- **Verify State Transitions:** Check that the FSM transitions between states according to the defined rules and conditions. For example, verify that the system correctly transitions from S1 to S2 after the timer expires, and from S2 to S3 when a vehicle is detected on the East-West side.
- **Output Behavior:** Ensure that the traffic light signals (Green, Yellow, Red) correspond to the correct states. For instance, during state S1, the North-South light should be green, and the East-West light should be red.

#### Boundary Testing:

- **Edge Cases:** Test edge cases such as transitions when the system is just about to transition from one state to another (e.g., near the end of the timer). This will help ensure that the system handles transitions smoothly and without errors.

#### Timing Analysis:

- **Check Timings:** Verify that the FSM operates within the desired timing constraints. For example, ensure that each signal phase lasts for the expected time, and there are no delays or overlaps.

## 5. RESULTS AND EVALUATION

The FSM-based traffic signal control system presents several benefits in terms of efficiency, simplicity, and reliability. However, it also has some limitations, particularly when it comes to adaptability in dynamic traffic conditions. Below is an analysis of the effectiveness of this approach, followed by a discussion of potential drawbacks.

### 5.1. Effectiveness of FSM-Based Traffic Signal Control

#### Efficiency:

- **Predictable Timing:** One of the major strengths of the FSM-based traffic signal control system is its **predictability**. Since the system follows a well-defined set of states and transitions based on time or simple inputs, it ensures that the traffic light phases are consistent and reliable. This can reduce congestion at intersections during normal traffic flow and improve traffic management efficiency.
- **Low Computational Overhead:** The FSM is a **simple and low-complexity system**, which does not require intensive computation. It operates based on a fixed set of rules, making it computationally efficient and inexpensive to implement, especially for smaller intersections or cities with limited resources.
- **Reduced Accidents and Confusion:** The clear and deterministic state transitions reduce the chances of **confusion** at intersections, which in turn decreases the likelihood of accidents. The well-structured cycle of green, yellow, and red phases ensures a predictable flow of traffic and enhances road safety.

### **Simplicity:**

**Easy to Implement:** The design of the FSM-based controller is relatively simple compared to more complex adaptive systems. The use of basic **logic gates** and **flip-flops** to implement state transitions makes it easy to design, simulate, and implement in hardware or software environments like **Logisim**.

- **Minimal Maintenance:** The deterministic nature of FSMs means the system doesn't require continuous monitoring or real-time adjustments, simplifying maintenance and reducing the likelihood of failure due to errors or unforeseen conditions.
- **Cost-Effective:** FSM-based systems are highly cost-effective, as they use basic components (e.g., flip-flops, logic gates) which are inexpensive and easy to maintain. This makes FSM-based traffic signal control a good option for cities or areas with limited budgets.

## **5.2. Potential Drawbacks and Limitations**

### **Lack of Adaptability in Dynamic Conditions:**

- **Fixed Time Phases:** One of the most significant drawbacks of an FSM-based traffic signal control system is its lack of **adaptability**. Since the FSM follows predefined time intervals for each phase, it cannot adjust dynamically to fluctuations in traffic volume or congestion. For instance, during rush hours, the system may fail to allocate longer green light phases to heavily trafficked lanes or reduce green time on less busy lanes, leading to **inefficiency**.
- **No Real-Time Adjustment:** FSMs are not inherently capable of responding to real-time data (e.g., vehicle presence, traffic speed, or pedestrian crossings). A static FSM system cannot prioritize certain vehicles, such as **emergency vehicles**, or adapt to unusual traffic conditions, such as a **traffic accident** or a **construction zone**. As a result, the system may either over or underutilize the available time for green lights in certain directions.

### **Scalability:**

- **Limited Scalability:** FSM-based systems are typically effective for simple intersections with straightforward traffic patterns. However, as the complexity of the traffic network increases, such as with multiple lanes, turning signals, or complex intersections, the FSM design can become more complicated. Managing the interactions of various traffic flows with an FSM becomes harder, especially when the number of states and transitions increases, making the system less scalable.

### **Lack of Traffic Sensing Integration:**

- **Fixed Signal Timing:** While FSMs can be enhanced with **external sensors**, such as inductive loops or cameras, to detect vehicle presence, this requires additional hardware and can increase the complexity of the system. Without such sensors, FSMs are purely time-based and cannot adjust based on the actual number of vehicles at the intersection. This lack of **real-time traffic sensing** means that the system could either be inefficient during peak traffic times or underutilized during off-peak periods.

### **Complexity in Multi-Intersection Coordination:**

- **Coordination Issues:** In larger cities or high-traffic areas, traffic signal systems need to be coordinated across multiple intersections to avoid congestion and ensure smooth traffic flow. FSM-based controllers, while effective at a single intersection, are not inherently designed to coordinate traffic lights across multiple intersections without additional logic or communication between signals. This can lead to **inefficiencies** in traffic flow, as signals may not be synchronized or may cause "bunching" of cars when moving from one intersection to another.

### 5.3. Comparison with Other Control Strategies

Compared to **timer-based** or **sensor-based control** systems, FSM-based traffic signal controllers are simpler but less flexible. **Timer-based control** is similar in simplicity, but **sensor-based systems** or **adaptive control algorithms** are far more efficient in handling real-time traffic flow changes. Adaptive systems, such as those using **machine learning** or **reinforcement learning**, can dynamically adjust signal phases based on traffic density and patterns. These systems are much more complex to implement but offer significant improvements in managing **highly variable traffic conditions**.

### 5.4. Enhancements for FSM-Based Systems

To overcome some of the limitations of FSM-based control systems, several improvements can be made:

- **Integrating Traffic Sensors:** By incorporating traffic sensors at key points, FSMs can adjust signal timings based on vehicle count, allowing for more responsive control. This would bring the FSM closer to the performance of adaptive systems.
- **Coordinating with Traffic Management Systems:** FSMs could be integrated with **centralized traffic management systems** that communicate between multiple intersections. This would enable the coordination of signal phases across a broader area, improving overall traffic flow and reducing congestion.
- **Hybrid Systems:** A hybrid approach could combine FSMs with adaptive systems, where FSMs handle basic control and adaptive algorithms adjust signal timing for specific scenarios. This would allow for more dynamic control while maintaining the simplicity and reliability of FSMs.

## 6. CONCLUSIONS

This paper explored the design and implementation of a traffic signal control system using **Finite State Machines (FSMs)** and discrete logic circuits. Through this approach, we demonstrated how FSMs can efficiently model traffic signal states and transitions, ensuring a structured, predictable, and safe operation for traffic flow management at intersections.

### Key Findings

- **FSM-based Traffic Control:** The FSM model provides a robust and deterministic method for controlling traffic signals. By encoding different signal phases (Green, Yellow, Red) as states, FSMs facilitate clear and controlled transitions based on inputs like timers and vehicle sensors. This ensures that traffic lights follow a predictable pattern, reducing the chances of accidents and improving the overall safety of road users.
- **Role of Discrete Logic:** The paper highlighted the role of basic digital components, such as **AND**, **OR**, and **NOT** gates, in implementing the FSM-based traffic signal system. By using these gates, we can create the state transition logic and outputs that control the traffic lights based on the current state and external inputs like sensors or timers. The simplicity of these components makes the FSM design cost-effective, reliable, and easy to implement, especially in environments with limited budgets or technical infrastructure.
- **Cost-Effectiveness and Reliability:** FSMs, when implemented with discrete logic, provide a low-cost and highly reliable solution for traffic signal control. This makes them ideal for cities with fewer resources or for smaller intersections where advanced systems may be overkill. The straightforward nature of FSMs reduces the risk of malfunction and ensures that the system remains predictable and easy to maintain.

- **Limitations of Static FSMs:** While FSMs work well for simple, pre-programmed traffic control, they do not adapt to changing traffic conditions in real-time. In environments with dynamic traffic flow or during peak hours, a fixed FSM-based system may become inefficient. This limitation highlights the need for enhancements that allow the system to respond flexibly to real-time traffic data.

### **Possible Enhancements**

- **Incorporating Adaptive Algorithms:** One significant enhancement is the integration of **adaptive algorithms** that can adjust signal timings based on real-time traffic conditions. For instance, **machine learning** or **reinforcement learning** algorithms can optimize traffic light phases by learning from traffic patterns and adjusting the signal timings accordingly. This would reduce congestion during peak traffic periods and improve overall traffic flow.
- **Advanced Sensor Integration:** Another improvement would be the use of more sophisticated sensors, such as **LiDAR**, **radar**, or **camera-based systems**, to provide more detailed data about traffic conditions. These sensors can detect not only vehicle presence but also speed, type of vehicle, and pedestrian movement. This data could be used to dynamically adjust signal timings, prioritize emergency vehicles, or optimize traffic flow based on the number of vehicles and their speed.
- **Vehicle-to-Infrastructure (V2I) Communication:** Integration of **V2I communication** could enhance the system by allowing vehicles to communicate with traffic signals. Vehicles equipped with V2I technology could inform signals about their location, speed, and intent, allowing the traffic lights to prioritize certain lanes, improve coordination between intersections, or give priority to buses and emergency vehicles. This would enhance the efficiency and responsiveness of the traffic control system.
- **Smart City Integration:** FSMs could also be integrated into broader **smart city** networks, allowing traffic signals to work in coordination with other urban systems. By linking traffic signal control with public transportation systems, emergency services, and pedestrian management, we could dynamically adjust traffic signals to optimize city-wide transportation flow, reducing congestion and enhancing safety.

### **BOOKS AND TEXTS:**

**Kuo,F. (2010).**

*Digital Control Systems: Design, Identification, and Implementation.*

John Wiley & Sons.

- This textbook covers the fundamentals of digital control systems, including FSMs, which are widely used in traffic signal control systems.

**Mendel,J.M. (2001).**

*Discrete Event Systems: Analysis and Control.*

Prentice-Hall.

- A comprehensive textbook on the analysis and control of discrete event systems, with applications to traffic control.

### **WEBSITES AND RESOURCES:**

- **IEEE Xplore Digital Library**

<https://ieeexplore.ieee.org/>

- An excellent resource for accessing research papers and articles related to FSMs, traffic control, and other engineering topics.

- **Science Direct**

<https://www.sciencedirect.com/>

- Another valuable resource for academic papers, including those on traffic signal control and FSM applications.

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